# Five Million Parameters on CIFAR-10: What Worked, What Did Not, and an Unexplained Test Gap

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**Abstract**

We train a 4.88M-parameter ResNet from scratch on CIFAR-10 under a hard 5M parameter budget and a hidden evaluation set. Our best recipe combines a long cosine schedule, SAM, EMA, and knowledge distillation from three auxiliary teachers, reaching 97.84% clean validation accuracy (97.92% with 10-view TTA). The team's selected submissions scored 0.911 and 0.904 public; the private score was 0.880, placing 13th. The roughly ten-point gap between validation and private accuracy is the central unsolved problem: the unlabeled images released with the starter data were not the final evaluation images, so our shift analysis is exploratory. We also report what did *not* work: RandAugment, AutoAugment-CIFAR, geometry-only and affine+color augmentation, ASAM/SWA fine-tuning, logit-standardized KD, a two-model parameter split, multi-crop TTA, test-time BN adaptation, Sinkhorn balancing, and seed ensembling. One-view AugMix was the one clean-neutral augmentation that consistently improved corrupted and photometric proxies, but it was not shipped because its KD confirmation finished after submission selection. We argue that negative results and explicit uncertainty about the test distribution are as useful as the final score.

## 1 Introduction

CIFAR-10 is a small-data benchmark that modern architectures have largely saturated: with tens of millions of parameters and long schedules, test accuracy above 99% is routine. The problem changes when the model must be trained from scratch under a hard budget of five million trainable parameters, and when the evaluation set is hidden and may not match the training distribution. Both constraints were active in this competition. The first forces careful architecture and optimization choices; the second makes validation accuracy an imperfect guide and turns the leaderboard into a noisy, partially observed signal.

We approached the problem as a two-track collaboration. One track pushed clean accuracy through architecture search, long schedules, and knowledge distillation (KD) from large teachers trained on the same data. The other track studied robustness: it characterized the only unlabeled images available to us, built corrupted and photometric proxy validation sets from the training split, and ran controlled augmentation ablations. The two tracks shared one codebase, one fixed 5,000-image holdout, and one set of conventions, so every number in this paper is directly comparable.

The main result is a 4.88M-parameter student that reaches 97.84% clean validation accuracy and 97.92% with 10-view test-time augmentation (TTA). The team's selected submissions scored 0.911 and 0.904 on the public leaderboard; the team's private score was 0.880, placing 13th. The roughly ten-point gap between our best holdout number and the private score is the central unsolved problem of the project.

An important caveat shapes how we interpret that gap. The 2,000 unlabeled images distributed with the starter data, which we analyzed at length, were not the images used for the final leaderboard. Our photometric statistics, nearest-neighbor tests, and visual comparisons therefore describe that released sample, not the final test set. They motivated the robustness experiments and gave us a useful analysis pipeline, but they cannot explain the final score. We report them as exploratory evidence and explicitly separate them from the leaderboard outcome. The only direct evidence of a distribution difference is the gap between validation and leaderboard accuracy itself.

This paper makes four contributions.

- A strong, reproducible 4.88M-parameter recipe: long cosine training, SAM, EMA, TrivialAugment, Cutout, mixup/CutMix, and three-teacher KD. It improves a plain baseline from 94.76% to 97.84% validation (97.92% with TTA) and yields the team's 0.911 public / 0.880 private submissions.
- A controlled account of which ingredients mattered. Long schedules and KD gave the largest gains; architecture changes within the 5M budget gave small ones; several plausible tricks did not pay off.
- A clean-neutral augmentation result: one-view AugMix leaves clean accuracy unchanged but improves a composite corruption proxy by 2.2pp and a photographic-jitter proxy by 1.9pp, replicated across seeds and surviving distillation.
- A negative-results and uncertainty report. We document failed techniques with paired statistics and validation curves, and we separate proxy evidence from evidence about the hidden test set.

Section 2 reviews related work. Section 3 defines the protocol. Section 4 describes the architectures, training recipe, KD variants, augmentation policies, and proxy analyses. Section 5 presents accuracy and robustness results. Section 6

collects the techniques that underperformed. Section 7 discusses the gap and limitations; Section 8 concludes.

## 2 Related Work

**Efficient residual networks.** Residual connections (He et al. 2016) made very deep convolutional networks trainable. Wide Residual Networks (Zagoruyko and Komodakis 2016) showed that width is a more efficient axis than depth at a fixed parameter count, and PyramidNet (Han, Kim, and Kim 2017) grows channels gradually while using parameter-free shortcuts; ShakeDrop regularizes the resulting deep networks. Our students are a pre-activation bottleneck ResNet family with 4.88–4.92M parameters; the teacher ensemble uses WRN-28-10 and PyramidNet-110-270.

**Optimization and regularization.** Sharpness-Aware Minimization (SAM) (Foret et al. 2021) and its adaptive variant ASAM (Kwon et al. 2021) seek flat minima by perturbing weights before each update. Weight averaging, both EMA and SWA, is a cheap way to stabilize stochastic training. Cosine schedules with warmup and label smoothing are standard. We combine SAM, EMA, cosine decay, and label smoothing, and we test ASAM/SWA as late fine-tunes.

**Data augmentation.** TrivialAugment (Müller and Hutter 2021) samples a single operation and magnitude per image and is competitive without tuning. AutoAugment (Cubuk et al. 2019) and RandAugment (Cubuk et al. 2020) search or parameterize augmentation policies; Cutout (DeVries and Taylor 2017), mixup (Zhang et al. 2018), and CutMix (Yun et al. 2019) regularize by occluding or mixing images. AugMix (Hendrycks et al. 2020) mixes multiple augmentation chains and adds a Jensen-Shannon divergence consistency loss; in our implementation we use the image-only, one-view variant without JSD. We compare these policies at matched schedules and on clean and shifted proxies.

**Knowledge distillation.** KD (Hinton, Vinyals, and Dean 2015) trains a small student on the softened outputs of one or more teachers. Long schedules and consistent teacher/student views are important for state-of-the-art distillation. Logit-standardized KD (LS-KD) (Sun et al. 2024) removes per-example scale and shift before matching distributions and reports better calibration. We use ordinary KL distillation with two or three teachers and evaluate LS-KD as a matched alternative.

**Distribution shift and test-time adaptation.** CIFAR-10.1 (Recht et al. 2018) showed that natural distribution shifts can produce large accuracy drops even for models that generalize well in-distribution. The common-corruption benchmark (Hendrycks and Dietterich 2019) provides synthetic stress tests; covariate-shift adaptation attempts to close the gap without labels. Sinkhorn balancing is a related class-prior correction. We use corrupted and photometric proxies as stress tests, and we evaluate BN adaptation and Sinkhorn balancing as negative results. Crucially, the unlabeled sample available to us was not the final evaluation set, so we do not claim that any of these proxies models the actual shift.

## 3 Setup and Protocol

**Data and splits.** We use the 50,000 labeled CIFAR-10 training images (10 classes, $32 \times 32$ RGB). A fixed stratified 5,000-image holdout (split seed 0) is used by every student and every teacher; the remaining 45,000 images are used for training. Teachers and students must hold out the same images, otherwise teacher predictions on the holdout leak into the student through the soft targets. The final full-data models are retrained on all 50,000 images once the recipe is fixed. No external data and no pretrained weights are used.

**Evaluation.** The competition metric is top-1 accuracy. The public and private leaderboards each use 1,000 hidden images, so a public score near 90% has a standard error of about 1.1 percentage points; we do not select models by small public differences. Unless stated otherwise, validation numbers use EMA weights. We report raw, EMA, flip TTA (average of the image and its horizontal mirror), and 10-view TTA. The latter, which we call *5-crop* to match the competition codebase, averages the softmax over a center view and four $\pm 2$-pixel shifts, each with and without a horizontal flip.

**Paired statistics.** Model comparisons use McNemar's exact test on per-image correctness over the same 5,000 holdout images. We report the discordant counts and two-sided binomial $p$-values. The seed-to-seed spread on this holdout is about 0.2pp on clean data and 0.4pp on the composite proxy, so differences of that size are treated as noise.

**Rules.** The official overview asks for a modified ResNet with at most 5M trainable parameters, trained from scratch on the permitted CIFAR-10 training data, with no pretrained weights or external datasets. The evaluation images may not be labeled, used for training, or used for model selection. We apply the budget to the submitted network; the teachers are auxiliary models trained from scratch on the same 50,000 images and are not part of the submitted inference system. The team submits a single network per submission.

**The released unlabeled sample.** During the competition we had access to 2,000 unlabeled images distributed with the starter data. We later learned that these were *not* the images used for the final leaderboard. We nevertheless used them to build and sanity-check a shift-analysis pipeline: photometric statistics, nearest-neighbor matching against the 50k training set under dihedral symmetries and translation-tolerant pooling, and a visual montage. Those analyses describe the released sample only. They motivated the robustness experiments but cannot explain the final test score.

**Proxies.** We build shifted validation sets from the 5,000 holdout images only. The corruptions are Gaussian noise ($\sigma = 0.05, 0.10$), Gaussian blur (radius 1.0, 2.0), JPEG compression (quality 30), contrast scaling (0.7, 1.3), color saturation (1.3), and a composite *combo* (contrast 1.15, brightness 1.05, blur 0.5, Gaussian noise $\sigma = 0.02$, JPEG 75). The photographic proxies apply per-image brightness/contrast/saturation scaling plus a small zoom and rotation: *mild* uses $\pm 8\%$ brightness/contrast, $\pm 12\%$ saturation, zoom 1.05 and $\pm 3°$; *strong* uses $\pm 20\%$, $\pm 30\%$, zoom 1.15

and $\pm 6^\circ$. These are label-free stress tests, not models of the final shift.

**Compute.** Main runs used NYU HPC A100 nodes; ablations used A10-class cloud GPUs and a local RTX 2070. A 4.9M student takes about 19–21s per epoch on an A10-class GPU; three-teacher distillation takes about 59s per epoch. The longest runs (1,200 epochs on all 50k images) take several hours.

# 4 Methods

## Architectures

**Students.** Our student family is a pre-activation bottleneck ResNet. A $3 \times 3$ stem produces the first stage's channels and keeps the $32 \times 32$ resolution; three stages then use 3/5/3 or 2/3/3 basic blocks, with the first block of stages 2 and 3 applying stride 2. Each block computes $\mathrm{ReLU}(F(x) + S(x))$, where $F$ is conv–BN–ReLU–conv–BN, the last BN is zero-initialized so the block starts as an identity, and $S$ is an identity or a $1 \times 1$ conv–BN shortcut when the shape changes. A global average pool and a linear layer produce the logits. The two selected presets are `r3_64_3-5-3` (64/128/256 channels, 3/5/3 blocks, 4.92M parameters) and `r3_80_2-3-3` (80/160/256, 2/3/3, 4.88M). A 300-epoch sweep also evaluated `r3_96_2-2-2` (4.90M), `r4_64_3-3-2-1` (4.89M), and `r3_40_8-8-8` (4.72M).

**Teachers.** We use two WRN-28-10 teachers (Zagoruyko and Komodakis 2016) (36.5M parameters each, seeds 0 and 1) and one PyramidNet-110-270 teacher (Han, Kim, and Kim 2017) with ShakeDrop (28.5M, seed 2). Teachers are trained for 400 epochs with the same augmentation recipe but a different seed each; teacher size is not constrained by the 5M budget because only the student is submitted. The teacher pool reaches 98.14% holdout accuracy (WRN seeds 0+1); the individual teachers reach 98.06, 98.12, and 97.64.

## Training Recipe

All students use SGD with Nesterov momentum 0.9, batch size 128, base learning rate 0.1, weight decay $5 \times 10^{-4}$ (not applied to BN affine parameters or biases), a 5-epoch linear warmup followed by cosine decay to zero, and bf16 autocast. The default augmentation is random crop with 4-pixel padding plus horizontal flip, TrivialAugment (Müller and Hutter 2021), Cutout 16 (DeVries and Taylor 2017), and per-batch mixup (Zhang et al. 2018) ($\alpha = 0.2$) or CutMix (Yun et al. 2019) ($\alpha = 1.0$) with probability 1.0. Mixing is disabled for the last 15 epochs of 300-epoch runs, 30 epochs of 600-epoch runs, and 20 epochs of teacher runs. Label smoothing is 0.1 when training from hard labels and is disabled when KD is on.

SAM (Foret et al. 2021) with $\rho = 0.05$ costs two forward/backward passes per step; the reported SAM runs use it for the whole schedule. EMA with decay 0.999 is maintained alongside the raw weights, and both are evaluated every epoch. The 300-epoch ablation schedule is the shipping schedule for the augmentation comparison; the main students use 600 or 1,200 epochs.

## Knowledge Distillation

For ordinary KD, each teacher sees the same augmented and mixed batch as the student, and its softened distribution is computed at temperature $T = 4$. The teacher distributions are averaged, and the student minimizes

$$\mathcal{L} = 0.9\, T^2\, \mathrm{KL}(\bar{p}_T \,\|\, q_T) + 0.1\, \mathrm{CE}(q, y), \qquad (1)$$

where $\bar{p}_T$ is the averaged teacher distribution and $q_T$ the student distribution at the same temperature. The CE term uses the mixup/CutMix soft targets. The two-teacher and three-teacher runs use this loss with `r3_64`; the selected `r3_80` runs use the same loss with all three teachers.

We also implement logit-standardized KD (LS-KD, Sun et al. (2024)). Each teacher's logits are z-scored per example before softening, the ensemble is averaged in that space, and the student's logits are standardized identically. The loss is $0.1\,\mathrm{CE} + 9\,T^2\,\mathrm{KL}$ with $T = 2$. We evaluate LS-KD both as a 60-epoch continuation and as a fresh 300-epoch run.

## Augmentation Policies

We compare five alternatives to the TrivialAugment control, all at the same 150-epoch pilot schedule (no SAM, no KD):

- **RandAugment** (Cubuk et al. 2020), two operations per image.
- **AutoAugment-CIFAR** (Cubuk et al. 2019), the learned CIFAR policy.
- **Geometry only**: mild random affine (translation, scale, rotation).
- **Extra affine+color**: stronger affine plus color jitter.
- **AugMix** (Hendrycks et al. 2020), the torchvision implementation with severity 3 and mixture width 3, one view and no JSD term. This is the image-only variant; it does not add the consistency loss.

The winner is then evaluated at 300 epochs with SAM, replicated with a second seed, and finally tested on top of three-teacher KD. This schedule matters: the 150-epoch pilot overstates the augmentation effect because the 300-epoch control also improves substantially.

## Shift Analysis and Proxies

The released unlabeled sample is analyzed with simple photometric statistics (brightness, contrast, saturation, sharpness, high-frequency ratio, JPEG blockiness, and colorfulness), a nearest-neighbor test, and a visual montage. The nearest-neighbor test matches each sample image to all 50,000 training images under the 8 dihedral symmetries, and separately under 5 small translations with $8 \times 8$ average pooling. We compare the resulting distance distribution to that of genuine CIFAR-10 pairs (holdout vs. train) and report the fraction of sample images below the 1st percentile of the genuine distribution. A transformed CIFAR image would collapse toward zero under one of these matchings; genuinely different content should not.

The corrupted and photographic proxies described in Section 3 are built from the holdout only. They are used as stress tests and as a way to detect whether an augmentation improves robustness along a particular axis. They are not used

to select the final submission, and they are not claimed to model the final hidden set.

# 5 Results

## Main Accuracy

Table 1 summarizes the accuracy progression, and Figure 2(a) shows the corresponding validation curves. A plain 200-epoch student reaches 94.76% validation accuracy. Adding the full training recipe (300 epochs, TrivialAugment, Cutout, mixup/CutMix, SAM, EMA) raises this to 97.36%, a gain of 2.6 points. Two-teacher KD adds 0.38 points (97.74%), and a third teacher adds only 0.02 points (97.76%), which is within noise. Moving to the wider `r3_80` student and 600 epochs reaches 97.84% raw and 97.92% under 5-crop TTA. Extending to 1,200 epochs changes the holdout number by $-0.02$ raw and $+0.02$ under 5-crop (97.82/97.94), i.e. it is not distinguishable from 600 epochs on this split.

The full-data models have no unbiased validation score. The public leaderboard scores are 0.904 for the two-teacher student and 0.911 for the selected `F2s1` model (5-crop TTA; 0.903 with flip TTA only). The team's private score is 0.880, placing 13th. Because both `F2s1` and the two-teacher student were selected as final submissions, the private score cannot be attributed to a single model.

A 4.88M student beats the 28.5M PyramidNet teacher (97.64%) on the holdout, and comes within 0.3 points of the best teacher pool (98.14%). This is the expected effect of distillation with a long schedule and strong augmentation: the student inherits the ensemble's function while seeing more effective regularization than any single teacher.

## Architecture and Distillation Ablations

At 300 epochs and with three teachers, the architecture sweep gives 97.38% for `r3_64`, 97.42% for `r3_96`, and 97.62% for `r3_80`. The wider first stage is worth about 0.24 points over the baseline shape; the deeper/narrower alternatives do not help. We therefore use `r3_80` for the main runs.

The KD hyperparameter sweep is similarly flat. With `r3_80` at 300 epochs, temperature 2 reaches 97.62% and temperature 1 with $\alpha = 1$ reaches 97.50%, both no better than the $T = 4$ baseline (97.62%). Two teachers reach 97.74% and three reach 97.76% at 600 epochs; the third teacher adds 0.02 points, which is smaller than the seed spread. We keep three teachers for the main runs because the cost is modest and the ensemble is more stable, but we do not claim a real accuracy gain from the third teacher.

## Robustness and Augmentation

**The 150-epoch pilot.** Figure 1 compares six policies at the same cheap schedule (150 epochs, no SAM, no KD). The control (TrivialAugment) reaches 96.48% clean TTA and 84.32% on the composite proxy. RandAugment (96.50/83.70) and AutoAugment-CIFAR (96.62/83.14) are slightly better on clean data and worse on the proxy. Geometry-only (95.98/84.96) loses 0.50 clean for 0.64 proxy. Affine+color (95.46/84.38) loses clean accuracy without a proxy gain. One-view AugMix is the only policy that is clean-neutral (96.52, +0.04) and substantially better on the proxy (87.42, +3.10). The pilot motivated the full 300-epoch test, but its effect size is inflated: the 300-epoch control also improves a lot.

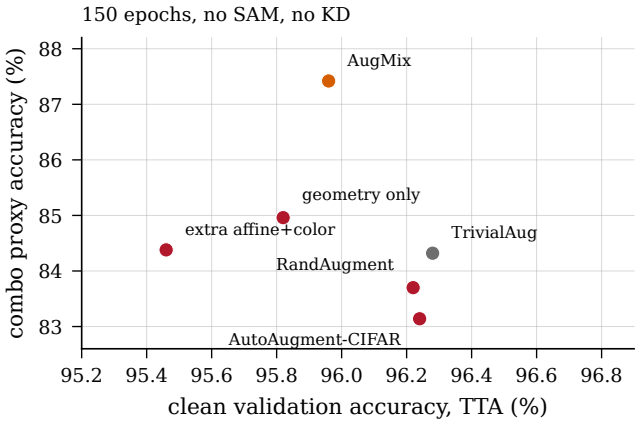



Figure 1: Augmentation pilot at 150 epochs (no SAM, no KD). Each point is one policy; the vertical axis is the composite corruption proxy. AugMix is the only policy that is clean-neutral and improves the proxy. The pilot overstates the effect because the 300-epoch control is stronger.

**The 300-epoch test.** At 300 epochs with SAM, AugMix and the TrivialAugment control are identical on clean validation (EMA 97.08% both; flip TTA 97.04 vs. 97.22). Under the no-TTA shifted evaluation, AugMix improves the composite by 2.18 points (89.02 vs. 86.84, McNemar $p = 3.6 \times 10^{-9}$), Gaussian noise 0.05 by 4.30, and JPEG-30 by 5.16. Under flip TTA it improves mild and strong photographic jitter by 1.10 and 1.38 points ($p \approx 10^{-5}$). Geometry alone is where AugMix does nothing (+0.06 on $\pm 3$-pixel shifts), which is consistent with the gain coming from photometric variation rather than framing. A second AugMix seed reproduces the effect (+1.80 composite; mean +1.99 across seeds) and keeps clean accuracy neutral (96.90%).

**Distillation.** The provisional KD check uses the same `r3_80` student, three teachers, and 300 epochs, changing only `--autoaug`. In the checkpoint comparison at epoch 200 of 300 (the archived logs stop earlier), AugMix is clean-neutral ($-0.12$, $p = 0.56$) and better on the composite (+2.20, $p = 3.7 \times 10^{-12}$) and strong photographic jitter (+1.92, $p = 5.6 \times 10^{-12}$); the geometry proxy moves by +0.36 ($p = 0.069$). At that point the AugMix arm is level with the 600-epoch A4 model on the composite (89.26 vs. 89.20) while still about one point behind on clean (96.86 vs. 97.84). The effect therefore provisionally survives distillation at roughly the same size as in the no-KD comparison. The run was interrupted and the two logs stop at epochs 163 and 178 of 300; the comparison is provisional and was not used for submission selection.

**The JSD caveat.** The B4/B5 variant is image-only, one-view AugMix without the Jensen-Shannon consistency

Table 1: Main results on the fixed 5,000-image holdout and the leaderboard. "Val" is the best reported validation accuracy (raw for baseline, EMA for KD/A4/V2, EMA+flip TTA for all tricks); "5-crop" is the 10-view TTA described in Section 3. Full-data models are trained on all 50k images and have no honest validation score. The private 0.880 is the team's best of the two selected submissions; both were selected, so the private 0.880 (13th) is not attributed to a single row.

| Run | Architecture | Params | Epochs | Val | 5-crop | Public / Private |
|---|---|---|---|---|---|---|
| Baseline | `r3_64_3-5-3` | 4.92M | 200 | 94.76 | – | – |
| All tricks, no KD | `r3_64_3-5-3` | 4.92M | 300 | 97.36 | – | – |
| KD, 2 teachers | `r3_64_3-5-3` | 4.92M | 600 | 97.74 | – | 0.904 / – |
| KD, 3 teachers | `r3_64_3-5-3` | 4.92M | 600 | 97.76 | – | – |
| A4 (3 teachers) | `r3_80_2-3-3` | 4.88M | 600 | 97.84 | 97.92 | – |
| V2 (3 teachers) | `r3_80_2-3-3` | 4.88M | 1200 | 97.82 | 97.94 | – |
| F1 / F2 (full data) | `r3_80_2-3-3` | 4.88M | 1200 | – | – | 0.898 / 0.899 |
| **F2s1 (selected)** | `r3_80_2-3-3` | 4.88M | 1200 | – | – | **0.911 / –** |

term. A separate 80-epoch AugMix+JSD finisher (JSD weight 12, ramped over 10 epochs) behaved differently: it gained 2.2 points on the blur proxy relative to the matched control but lost about one point on clean validation (96.88 vs. 97.44 for the control and 97.84 for untouched A4). It failed our clean-accuracy gate and was not submitted. The clean-neutral result should therefore be read as specific to the one-view variant at the schedules we tested, not to AugMix with JSD.

### Test-Time Augmentation

Flip TTA is the default. The 10-view 5-crop procedure adds essentially nothing on clean data (trivial $-0.02$, AugMix $+0.10$) and is a wash overall: it helps on JPEG and the composite by 0.5–1.2 points, but loses on noise and zoom. It costs ten times the inference. We nevertheless used 5-crop for the 0.911 submission because it exactly reproduced the team's accepted CSV; the gain over flip is inside leaderboard noise and should not be attributed to TTA.

### What the Released Sample Showed

The 2,000 unlabeled images released with the starter data are visibly unlike CIFAR-10: they are full-frame photographs with different framing, backgrounds, and object scale (Figure 4). On this sample, 0/2,000 are exact duplicates of a training image. The photometric differences are small (sharpness Cohen's $d = 0.27$, contrast $d = 0.17$, brightness $d = 0.14$, blockiness $d = -0.07$), and a nearest-neighbor test finds that the sample sits at or above the distance distribution of genuine CIFAR-10 pairs: median 1,982 vs. 1,852 under dihedral matching, and 79.9 vs. 73.5 under translation-tolerant pooling; 0/2,000 fall below the 1st percentile in either test. A brightness/contrast change, re-JPEG, flip/rotation, or shift-and-crop of a CIFAR image would collapse that residual toward zero; this sample does not.

These observations describe the released sample, not the final evaluation set. They motivated the hypothesis that the hidden set contained natural photographs, and therefore the decision to prioritize photometric robustness, but the hypothesis was never confirmed on the final images. The final leaderboard gap is the only direct evidence of a difference: our best holdout model is at 97.94% (5-crop), the selected submissions scored 0.911 and 0.904 public, and the team's private score is 0.880. The private score is about ten points below the holdout number. We cannot say whether that gap is caused by new photographs, a different capture pipeline, a harder class mix, or a combination; the released sample cannot adjudicate between them.

## 6 What Did Not Work

The competition rewarded clean accuracy, but a large fraction of our experiments did not improve it. We report them here because the failure pattern is informative: several techniques that are strong in the literature are neutral or negative at this budget and schedule, and several apparent gains disappear under a matched control. Table 2 summarizes the results; the text gives the context.

**Augmentation.** RandAugment and AutoAugment-CIFAR are clean-neutral or slightly positive, but both make the composite proxy worse than the TrivialAugment control. Geometry-only augmentation trades 0.50 clean points for 0.64 proxy points, which fails our clean-drop budget; affine+color loses 1.02 clean points with no proxy gain. AugMix is the exception and is discussed in Section 5. The lesson is that a policy can look competitive on clean validation and still be the wrong choice for robustness; we therefore require both clean neutrality and a proxy gain before adopting an augmentation.

**ASAM and SWA.** Figure 2(c) shows the five 60-epoch continuations from A4; none ends above the untouched start. A 60-epoch ASAM continuation beats the matched SAM continuation by 0.18 points (98.00 vs. 97.82, $p = 0.049$), but it does not beat the untouched A4 start point (97.92; $+0.08$, $p = 0.50$). The confirm run from the two-teacher student moves by +1 image against its SAM control and +2 against the untouched model ($p = 0.83$); pooled over both start points, ASAM versus the untouched model is +6 of 10,000 images, but the two comparisons reuse the same 5,000 images, so this pooled count is exploratory. ASAM is better than SAM for this late continuation, but the continuation itself does not improve the model. SWA is neutral (97.80 vs. 97.82). A fresh 300-epoch ASAM run changes one image (97.64 vs. 97.62); applying the 60-epoch ASAM finish to F2s1 changes 33/2,000 test labels and scores 0.908 public versus 0.911 for the untouched model; the correspond-

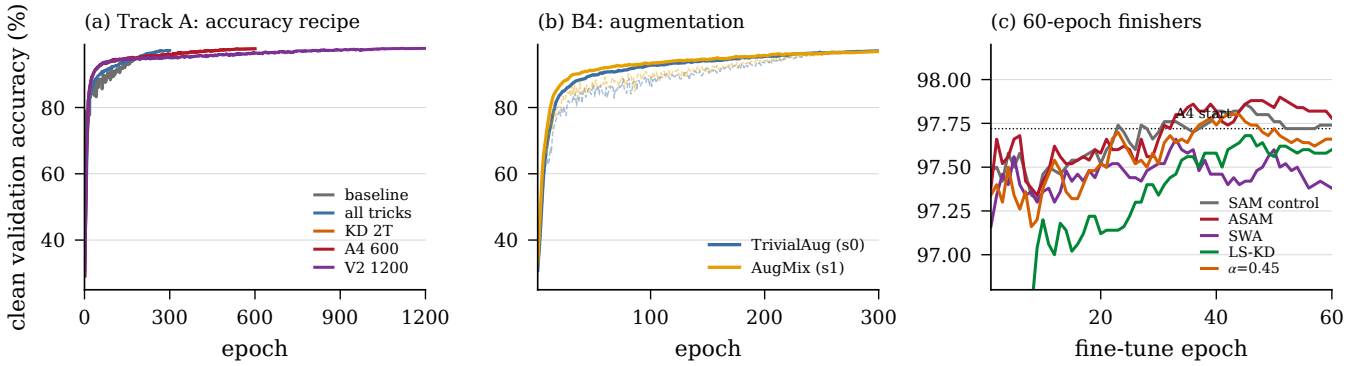



Figure 2: Validation curves (EMA weights unless noted, no TTA). (a) Track A: baseline (raw), all tricks, two-teacher KD, A4 (600 epochs), and V2 (1,200 epochs). (b) B4 augmentation at 300 epochs; the AugMix curve is seed 1 because the seed-0 log was not preserved (final EMA 97.08 vs. 96.88 for seed 1), and raw accuracy is dashed. (c) 60-epoch continuations from A4; the dotted line is the untouched A4 start (97.72 EMA, no TTA). Every continuation ends at or below that start. Final flip/5-crop TTA numbers are in Tables 1 and 2.

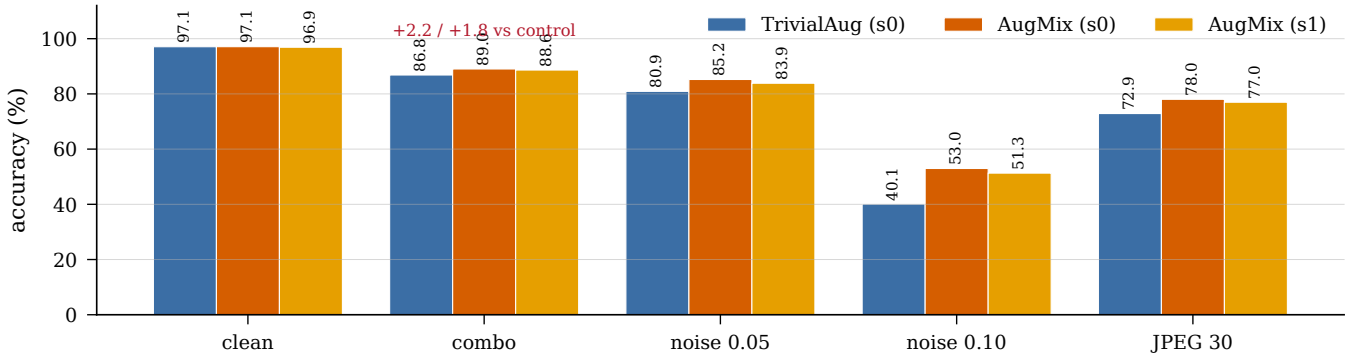



Figure 3: Clean and shifted accuracy for the 300-epoch B4 arms (EMA weights, no TTA). One-view AugMix is clean-neutral and improves the composite, noise, and JPEG proxies; the two seeds give +2.18 and +1.80 on the composite. These are label-free stress tests, not measurements of the final test set.

ing weight soup also scores 0.908. We did not launch fresh 1,200-epoch ASAM/SWA runs.

**LS-KD.** Logit-standardized KD is a calibration intervention in our hands, not an accuracy one. The continuation matches ordinary KD (97.76 vs. 97.82, within noise) and the fresh 300-epoch run matches its ordinary-KD baseline exactly (97.62), while substantially lowering validation NLL (0.086 vs. 0.154). If the metric rewarded calibration, LS-KD would be attractive; for top-1 accuracy it is a wash.

**Splitting the budget.** Two WRN-40-2 models (2.24M each, 4.49M together) reach 97.10 and 97.08 individually; their equal-probability ensemble reaches 97.26. The ensemble gains only +0.16 over one WRN and remains 0.36 below a single 4.88M `r3_80` (97.62). With this recipe, splitting the parameter budget into two smaller models is worse than spending it on one. The pair's errors are also highly correlated (111 images wrong in both; Jaccard overlap 0.61), so the ensemble has little diversity to exploit.

**TTA, adaptation, and ensembling.** The 10-view TTA is a wash on clean data and only helps on JPEG/composite proxies; flip TTA captures most of it. Test-time BN adaptation, which recomputes BN statistics on the evaluation images, drops the public score from 0.899 to 0.884. Sinkhorn class balancing is neutral (0.898 vs. 0.899). A two-seed AugMix ensemble gains +0.34 on the composite ($p = 0.15$) and +0.44 on strong photographic jitter ($p = 0.02$), but the composite gain is not significant and both are inside the public-LB noise; we did not use an ensemble for the final submission. Train-split BN recalibration (recomputing BN on clean training images) hurts badly: 97.08 vs. 97.92 for A4 and 97.20 vs. 97.94 for V2. The saved BN statistics are already well matched to the evaluation transform; replacing them with clean-image statistics removes the augmentation-induced robustness.

**Fresh 300-epoch sweep.** A matched 300-epoch sweep found no gain from extra regularization or a different KD weight: two-teacher KD is neutral (97.66), KD $\alpha = 0.45$ loses 11 images, and stochastic depth 0.1 loses 18 (97.30 vs. 97.66, $p = 0.005$).

(a) CIFAR-10 training images (reference)



(b) released unlabeled sample (not the final evaluation set)

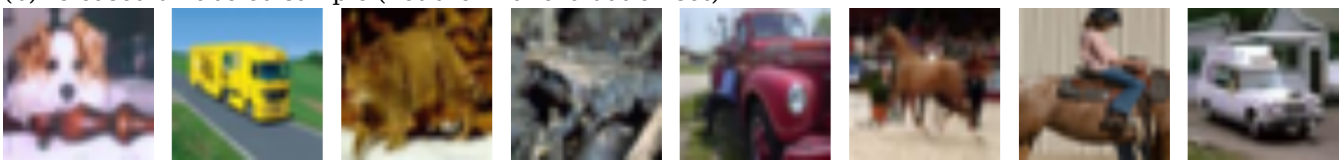


Figure 4: Top: CIFAR-10 training images. Bottom: the 2,000 unlabeled images released with the starter data. These are *not* the final evaluation images; they are shown only to document the sample that motivated our robustness experiments. The visual difference in framing and scale is why we hypothesized a natural-image shift, but the sample cannot explain the final leaderboard gap.

Table 2: Techniques that did not pay off. $\Delta$ is in percentage points on the 5,000-image holdout unless the row uses the public leaderboard. Paired McNemar $p$-values are shown where recorded; "–" means no paired test was logged. Pilot numbers are clean TTA / combo; the B4 comparison is in Table 1.

| Technique | Matched comparison | $\Delta$ | $p$ | Verdict |
|---|---|---|---|---|
| *Augmentation (150-epoch pilot, clean TTA / combo)* | | | | |
| RandAugment | 96.50 / 83.70 vs. 96.48 / 84.32 | $+0.02$ / $-0.62$ | – | worse proxy |
| AutoAugment-CIFAR | 96.62 / 83.14 vs. control | $+0.14$ / $-1.18$ | – | worse proxy |
| Geometry only | 95.98 / 84.96 vs. control | $-0.50$ / $+0.64$ | – | clean drop |
| Extra affine+color | 95.46 / 84.38 vs. control | $-1.02$ / $+0.06$ | – | clean drop |
| *Optimization and KD* | | | | |
| ASAM continuation | 98.00 vs. 97.82 (SAM) | $+0.18$ | 0.049 | no gain over A4 |
| LS-KD (300 ep) | 97.62 vs. 97.62 (ordinary KD) | 0.00 | – | lower NLL only |
| WRN-40-2 pair | 97.26 vs. 97.62 (one `r3_80`) | $-0.36$ | – | loses |
| *Inference and adaptation* | | | | |
| 5-crop TTA | 97.20 vs. 97.22 (trivial); 97.16 vs. 97.06 (AugMix) | $-0.02$ / $+0.10$ | 1.0 / 0.33 | wash |
| Test-time BN | 0.884 vs. 0.899 public | $-1.5$ | – | hurts |
| Sinkhorn balance | 0.898 vs. 0.899 public | $-0.1$ | – | no gain |
| 2-seed ensemble | +0.34 combo, +0.44 strong-photo | $+0.3$–$0.4$ | 0.15 / 0.02 | inside noise |
| Train-split BN | 97.08 / 97.20 vs. 97.92 / 97.94 | $-0.8$ / $-0.7$ | – | hurts |

**What we did not test.** DKD, structural teacher pruning, SE blocks, and fresh 1,200-epoch ASAM/SWA/LS-KD runs were left untested; they are the most promising follow-ups but did not fit the compute budget.

## 7 Discussion and Limitations

**What produced the accuracy.** The largest single jump came from the training recipe rather than the architecture: the baseline at 94.76% became 97.36% with long training, TrivialAugment, Cutout, mixup/CutMix, SAM, and EMA. KD added another 0.4 points, and the wider student added about 0.2. The 4.88M student ends up better than the 28.5M PyramidNet teacher and within 0.3 points of the best teacher pool. This is consistent with the view that a long, well-regularized student can absorb the teacher ensemble's function without inheriting its capacity. Architecture search inside the 5M budget was secondary: the best shape is only 0.24 points above the baseline shape at 300 epochs.

**The gap we could not explain.** Our best holdout number is 97.94% (5-crop); the selected submissions scored 0.911 and 0.904 public, and the team's private score was 0.880. The private score is roughly ten points below the holdout number, and the gap between the best public score (0.911) and the private score (0.880) is 3.1 points. The public/private split is 1,000 images each, so part of the 3.1-point difference is sampling noise; the rest could be a harder private subset, a different capture distribution, or both. We cannot separate these causes.

The 2,000 unlabeled images released with the starter data looked like natural photographs and were clearly unlike CIFAR-10. They motivated a natural-image shift hypothesis and a robustness program. They were not, however, the final evaluation images. This is the most important methodological lesson of the project: a proxy can be visually compelling and statistically well behaved and still be the wrong proxy. We should have verified what the released sample was be-

fore building a narrative around it. The analysis pipeline remains useful (it is reproducible and we would reuse it), but its conclusions do not transfer to the final test set.

**Why one-view AugMix was not shipped.** One-view AugMix is the one robustness intervention that survived every check we ran: it is clean-neutral at 300 epochs, improves the composite by 2.2 points, improves photographic jitter by 1.1–1.9 points, replicates across seeds, and keeps its effect under three-teacher KD. It was still not in the final submission. The KD confirmation finished after the team had selected its submissions; its logs stop at epoch 178/300, and the full-data 1,200-epoch AugMix run was never trained. Given that the final private gap is unexplained, we cannot claim that shipping AugMix would have improved the score. The honest conclusion is that it was the best-supported robustness candidate we had, and the submission schedule did not allow us to test it at the shipping scale.

**Limitations.** Most comparisons use a single seed; the exceptions are the two AugMix seeds and the ASAM confirm. The holdout is 5,000 images, so the seed-to-seed spread is 0.2–0.4 points and small differences should not be over-interpreted. The B5 KD comparison is provisional at 200/300 epochs and was interrupted twice. The corruption and photographic proxies are synthetic and label-free; they are stress tests, not models of the final shift. The public and private leaderboards are 1,000 images each, so the leaderboard itself is noisy. We did not run fresh 1,200-epoch ASAM, SWA, or LS-KD experiments, and we did not test DKD or structural teacher pruning. Finally, the full-data models have no unbiased validation score, so the final selection relies on the holdout models and the team's judgment.

**What we would do differently.** First, verify what any released test sample actually is before building a shift story around it. Second, ablate augmentation at the shipping schedule, not at a cheap pilot: the 150-epoch pilot overstated AugMix's effect because the 300-epoch control improved so much. Third, keep per-epoch logs for every run in the repository; several curves in this paper are missing because the logs stayed on a cluster or a cloud volume. Fourth, complete the KD robustness run before submission selection, so that a clean-neutral augmentation has a chance to ship. Fifth, treat the public leaderboard as a noisy estimate and select by paired holdout comparisons, which is what the team did.

## 8 Conclusion

We trained a 4.88M-parameter ResNet from scratch on CIFAR-10 and reached 97.84% clean validation accuracy (97.92% with 10-view TTA). The team's selected submissions scored 0.911 and 0.904 public, with a private score of 0.880 and 13th place. The recipe is ordinary in its ingredients—long cosine training, SAM, EMA, TrivialAugment, Cutout, mixup/CutMix, and three-teacher KD—but the combination is strong at this parameter budget, and the student beats its largest teacher.

The project also produced a useful catalog of negative results. RandAugment, AutoAugment-CIFAR, geometry-only and affine+color augmentation, ASAM/SWA fine-tuning, logit-standardized KD, a two-model parameter split, multi-crop TTA, test-time BN adaptation, Sinkhorn balancing, and seed ensembling all failed to improve the metric in our hands. One-view AugMix was the exception: clean-neutral and consistently better on corrupted and photometric proxies, but not shipped because its KD confirmation finished after submission selection.

The unresolved question is the roughly ten-point gap between clean validation and the private leaderboard. The unlabeled sample we analyzed was not the final evaluation set, so we can describe the gap but not explain it. We hope the negative results and the explicit separation between proxy evidence and leaderboard evidence are useful to others working under similar constraints.

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